



A Cole–Hopf Transformation Based Fourth-Order Multiple-Relaxation-Time Lattice Boltzmann Model for the Coupled Burgers' Equations

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Abstract

In this work, a Cole–Hopf transformation based fourth-order multiple-relaxation-time lattice Boltzmann (MRT-LB) model for d -dimensional coupled Burgers' equations is developed. We first adopt the Cole–Hopf transformation where an intermediate variable θ is introduced to eliminate the nonlinear convection terms in the Burgers' equations on the velocity $\mathbf{u} = (u_1, u_2, \dots, u_d)$. In this case, a diffusion equation on the variable θ can be obtained, and particularly, the velocity $\mathbf{u}$ in the coupled Burgers' equations is determined by the variable θ and its gradient term $\nabla\theta$. Then we develop a MRT-LB model for the d -dimensional transformed diffusion equation, and present the corresponding macroscopic finite-difference scheme. At the diffusive scaling, the macroscopic modified equation of the developed MRT-LB model is derived through the Maxwell iteration method. With the aid of the free parameters in the MRT-LB model, we find that not only the modified equation at fourth order can be obtained, but also the gradient term $\nabla\theta$ can be calculated locally by the non-equilibrium distribution function with a fourth-order accuracy, **this indicates that theoretically, the MRT-LB model for d -dimensional coupled Burgers' equations can achieve a fourth-order accuracy in space.** Finally, some simulations are conducted to test the MRT-LB model, and the numerical

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results show that the proposed MRT-LB model has a fourth-order convergence rate, which is consistent with our theoretical analysis.

Keywords d -dimensional coupled Burgers' equations · Cole–Hopf transformation · Multiple-relaxation-time lattice Boltzmann model

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1 Introduction

The Burgers' equation, as an important kind of basic partial differential equations (PDEs), can be used to describe some nonlinear physical phenomena, such as turbulence [35], shock wave [8, 52], solitons [9], gas dynamics [41], traffic flow [47], nonlinear speed of sound [20], to name but a few. In addition, the coupled Burgers' equations, instead of a single one, are usually adopted to describe some coupled transport phenomena. However, it should be noted that the coupled Burgers' equations are nonlinear, and usually, it is difficult to obtain their analytical solutions. Thus, it is desirable and necessary to develop some numerical methods for the (coupled) Burgers' equation(s). In the past years, two important kinds of numerical approaches have been developed for the (coupled) Burgers' equation(s). The first one is traditional macroscopic numerical methods, such as the finite-difference method [2, 39, 40], the finite-element method [50], and the finite-volume method [49]. The other one is the mesoscopic lattice Boltzmann (LB) method, which is not only a highly efficient second-order kinetic theory based approach for the fluid flow problems governed by the Navier–Stokes equations [31, 55, 57, 60], but also has been successfully extended to solve the (coupled) Burgers' equation(s) [7, 10, 11, 23, 28, 33, 36, 38, 51, 53, 58, 64]. Actually, the existing LB models for the (coupled) Burgers' equation(s) [7, 10, 11, 23, 28, 33, 36, 38, 51, 53, 58, 64] can be divided into two main kinds. The first one is the direct approach where the LB model is developed to solve the Burgers' equation [7, 10, 11, 23, 28, 33, 36, 38, 58, 64], however, this approach may suffer from numerical instability due to the existence of nonlinear and coupled convection term(s), and it is usually limited to the one- or two-dimensional Burgers' equation [7, 10, 11, 23, 28, 33, 36, 38, 58, 64]. The second one is the indirect approach, which is more universal and stable, and mainly focuses on the d -dimensional coupled Burgers' equations [51, 53]. In the second approach, the Cole–Hopf transformation [19] is used to eliminate the nonlinearity and coupling in the d -dimensional coupled Burgers' equations, and thus a simple diffusion equation on an intermediate variable θ is obtained. Then one can adopt some numerical methods to solve the transformed diffusion equation [1, 44–46, 65, 66]. Moreover, the velocity $\mathbf{u}$ in the coupled Burgers' equations is further determined by the variable θ and the gradient term $\nabla\theta$ which, in the framework of LB method, can be calculated conveniently by the first-order moment of the non-equilibrium distribution function with a second-order accuracy [53].

It should be noted that all the works mentioned above focus on the popular single-relaxation-time LB (SRT-LB) model. However, the SRT-LB model becomes unstable when the relaxation time is close to $1/2$. To overcome this problem, the advanced multiple-relaxation-time LB (MRT-LB) model [12, 13, 22] with some additional adjustable parameters (e.g., the relaxation parameters), is more general than the SRT-LB model, and can be adopted to achieve better numerical stability and/or accuracy through adjusting some free relaxation parameters [21, 37, 43, 48]. For this reason, Yu et al. [63] developed a second-order MRT-LB

model for the one-dimensional Burgers' equation. To our knowledge, however, these available LB models for the (coupled) Burgers' equation(s) are only of second-order accuracy, and it is still unclear whether a high-order LB model for the $d(\geq 1)$ -dimensional (coupled) Burgers' equation(s) can be obtained. Actually, we note that some works have been made to develop high-order LB models for the diffusion equations. For example, Suga [56] proposed a fourth-order SRT-LB model with the D1Q3 lattice structure for the one-dimensional diffusion equation, then Lin et al. [42] extended this work, and developed a sixth-order MRT-LB model. Recently, Chen et al. [16] considered the two-dimensional diffusion equation, and proposed a fourth-order MRT-LB model where the D2Q5 lattice structure is adopted. However, these high-order LB models are limited to the one- and two-dimensional diffusion equations, and there is no general high-order LB model for the d -dimensional diffusion equation. In addition, in the framework of the Cole–Hopf transformation based LB model for the coupled Burgers' equations, we also need to develop a high-order scheme for the gradient term $\nabla\theta$, besides the high-order LB model for the variable θ in the diffusion equation. It should also be noted that through the Chapman–Enskog analysis method [14], the Maxwell iteration method [34, 62], the direct Taylor expansion method [32, 59], the recurrence equations method [27, 30] or the equivalent equations method [24–26], one can show that the LB method is of second-order accuracy. Besides these asymptotic analysis approaches, Bellotti et al. [5] investigated the relation between the mesoscopic LB method and the classical macroscopic finite-difference method, and demonstrated that the mesoscopic LB model corresponds to a macroscopic multiple-level finite-difference scheme. They further derived the truncation errors and modified equations of the mesoscopic LB model via the macroscopic finite-difference scheme [3]. Recently, Chen et al. [18] extended the previous works [3, 5] to the general propagation LB model. From these works [3, 18], we know that for the macroscopic finite-difference scheme of the LB model, its modified equations at any order obtained by the Taylor expansion method are equivalent to those obtained by the Maxwell iteration method. Based on the this fact, we will consider the Maxwell iteration method, and propose a Cole–Hopf transformation based fourth-order MRT-LB model for the d -dimensional coupled Burgers' equations in this work.

The rest of this paper is organized as follows. In Sect. 2, we first present the d -dimensional coupled Burgers' equations and the simple diffusion equation based on the Cole–Hopf transformation. In Sect. 3, we first develop a unified framework of the MRT-LB model for the d -dimensional transformed diffusion equation and present the corresponding macroscopic finite-difference scheme. Then at the diffusive scaling, we derive the fourth-order modified equation of the MRT-LB model and the fifth-order expression of the distribution function through the Maxwell iteration method, and also the conditions of the fourth-order MRT-LB model for the d -dimensional coupled Burgers' equations. In Sect. 4, some simulations are carried out to test the accuracy of the developed MRT-LB model, and finally, some conclusions are given in Sect. 5.

2 The Cole–Hopf Transformation for the d -Dimensional Coupled Burgers' Equations

We now consider the following d -dimensional coupled Burgers' equations,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = \nu \nabla^2 \mathbf{u}, \quad (1)$$

which satisfy the following initial, boundary, and potential symmetry conditions,

$$\text{initial condition: } \mathbf{u}(\mathbf{x}, 0) = \psi(\mathbf{x}, 0), \quad \forall \mathbf{x} \in \Omega, \quad (2a)$$

$$\begin{cases} \text{homogeneous Dirichlet boundary condition [29,65]: } \mathbf{u}_i(\chi_i, t) = 0, & \chi_i \in \partial\Omega, \quad t > 0, \\ \text{periodic boundary condition: } \mathbf{u}(\mathbf{x} + \mathbf{L}, t) = \mathbf{u}(\mathbf{x}, t), & \mathbf{x} \in \partial\Omega, \quad t > 0, \end{cases} \quad (2b)$$

$$\text{potential symmetry condition: } \nabla \mathbf{u}(\mathbf{x}, t) = (\nabla \mathbf{u}(\mathbf{x}, t))^T, \quad \forall \mathbf{x} \in \Omega, \quad t > 0, \quad (2c)$$

where $\mathbf{u} = (u_1, u_2, \dots, u_d)$ is the velocity, which is dependent on both space $\mathbf{x} = (x_1, x_2, \dots, x_d)$ and time t , ν is the viscosity coefficient. The physical domain is defined as $\Omega = \prod_{i=1}^d (b_i - a_i)$ ($a_i, b_i \in \mathbb{R}$, $b_i > a_i$) with the boundary $\partial\Omega$, the boundary point $\chi_i = (x_1, \dots, x_{i-1}, \chi, x_{i+1}, \dots, x_d)$ with $\chi = a_i$ and $\chi = b_i$. The vector $\mathbf{L}$ denotes the length in the periodic directions, $\psi = (\psi_1, \psi_2, \dots, \psi_d)$ is the known d -dimensional vector function. For the d -dimensional coupled Burgers' equations, the Cole–Hopf transformation is given by

$$\mathbf{u}(\mathbf{x}, t) = -2\nu \frac{1}{\theta(\mathbf{x}, t)} \nabla \theta(\mathbf{x}, t), \quad \forall \mathbf{x} \in \Omega, \quad t > 0. \quad (3)$$

With the help of Eq. (3), the coupled Burgers' equations (1) can be reformulated as the following d -dimensional diffusion equation on the variable $\theta(\mathbf{x}, t)$,

$$\frac{\partial \theta}{\partial t} = \nu \nabla^2 \theta, \quad (4)$$

with the corresponding initial and boundary conditions [29, 65]

$$\begin{aligned} \text{initial condition: } \theta(\mathbf{x}, 0) &= \exp \left(- \frac{\sum_{i=1}^d \int_{a_i}^{x_i} \psi_i(x_1, \dots, x_{i-1}, s, x_{i+1}, \dots, x_d) ds}{2d\nu} \right), \\ \forall \mathbf{x} \in \Omega, \end{aligned} \quad (5a)$$

$$\begin{cases} \text{homogeneous Neumann boundary condition: } \mathbf{n}_w \cdot \nabla \theta(\mathbf{x}, t) = 0 \text{ or } \partial_{\chi_i} \theta(\chi_i, t) = 0, \\ \quad \mathbf{x} \in \partial\Omega, \quad t > 0, \\ \text{periodic boundary condition: } \theta(\mathbf{x} + \mathbf{L}, t) = \theta(\mathbf{x}, t), \quad \mathbf{x} \in \partial\Omega, \quad t > 0, \end{cases} \quad (5b)$$

where $\mathbf{n}_w$ represents the outer normal vector of the boundary $\partial\Omega$.

3 The Fourth-Order MRT-LB Model for the d -Dimensional Coupled Burgers' Equations

In this section, we will first develop a unified framework of the MRT-LB model for the d -dimensional transformed diffusion equation (4). Then based on the previous work [3], we will perform an analysis on the macroscopic finite-difference scheme of the MRT-LB model, and further derive the modified equation at fourth order through the Maxwell iteration method [18]. Finally, we will present some details on how to calculate the velocity $\mathbf{u}$ in the coupled Burgers' equations (1) with a fourth-order accuracy.

3.1 The MRT-LB Model for the d -Dimensional Diffusion Equation

The evolution of MRT-LB model for the d -dimensional diffusion equation (4) can be written as

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \Lambda_{ik} (f_k(\mathbf{x}, t) - f_k^{eq}(\mathbf{x}, t)), \quad i = 1, 2, \dots, q, \quad (6)$$

¹where $f_i(\mathbf{x}, t)$ and $f_i^{eq}(\mathbf{x}, t)$ are the distribution function and the equilibrium distribution at position $\mathbf{x}$ and time t , Δt is the time step. The collision matrix $\Lambda = \mathbf{M}^{-1} \mathbf{S} \mathbf{M}$ with $\mathbf{M}$ and $\mathbf{S}$ being the transformation matrix and the relaxation matrix, respectively, q denotes the number of discrete velocities in the DdQq (q discrete velocities in d -dimensional space) lattice structure. In this work, we consider $q = 2d^2 + 1$, the velocity set $\mathbf{c}$ (the discrete velocity $\mathbf{c}_i$ is the i_{th} column of $\mathbf{c}$), the set of polynomials of the transformation matrix $\mathbf{M}$ (here we denote the polynomial set as $\mathbf{p}_\mathbf{M}(X_1, X_2, \dots, X_d)$ for brevity), and the diagonal relaxation matrix $\mathbf{S}$ are given by (the case $d > 3$ can be found in Appendix A)

$$d = 1: \begin{cases} \mathbf{c} = (0, 1, -1)c, \\ \mathbf{p}_\mathbf{M}(X_1) = (1, X_1, X_1^2), \\ \mathbf{S} = \text{diag}(s_0, s_1, s_2), \end{cases} \quad (7a)$$

$$d = 2: \begin{cases} \mathbf{c} = \begin{pmatrix} 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \end{pmatrix} c, \\ \mathbf{p}_\mathbf{M}(X_1, X_2) = (1, X_1, X_2, X_1^2, X_2^2, X_1 X_2, X_1 X_2^2, X_1^2 X_2, X_1^2 X_2^2), \\ \mathbf{S} = \text{diag}(s_0, s_1, s_1, s_{21}, s_{21}, s_{22}, 1, 1, 1), \end{cases} \quad (7b)$$

$$d = 3: \begin{cases} \mathbf{c} = \begin{pmatrix} 0 & 1 & 0 & 0 & -1 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \end{pmatrix} c, \\ \mathbf{p}_\mathbf{M}(X_1, X_2, X_3) = (1, X_1, X_2, X_3, X_1^2, X_2^2, X_3^2, X_1 X_2, X_1 X_3, X_2 X_3, \\ X_1 X_2^2, X_1^2 X_2, X_1 X_3^2, X_1^2 X_3, X_2 X_3^2, X_2^2 X_3, X_1^2 X_2^2, X_1^2 X_3^2, X_2^2 X_3^2), \\ \mathbf{S} = \text{diag}(s_0, s_1, s_1, s_1, s_{21}, s_{21}, s_{21}, s_{22}, s_{22}, s_{22}, 1, 1, 1, 1, 1, 1, 1, 1, 1), \end{cases} \quad (7c)$$

where $c = \Delta x / \Delta t$ is the lattice speed with Δx being the lattice spacing, the element $\mathbf{M}_{ij}$ of the transformation matrix $\mathbf{M}$ is determined by the i_{th} element of the polynomial set $\mathbf{p}_\mathbf{M}$ with $X_k = \mathbf{c}_{kj}$ ($k = 1, 2, \dots, d$), and the relaxation parameters s_0, s_1, s_{21} , and s_{22} are located in the range of $(0, 2)$. In order to recover the diffusion equation (4) correctly from the MRT-LB model (6), the equilibrium distribution function f_i^{eq} should be designed as

$$f_i^{eq}(\mathbf{x}, t) = w_i \theta(\mathbf{x}, t), \quad (8)$$

where w_i is the weight coefficient, and for simplicity, here we assume that the weight coefficient w_i satisfies the following relations,

$$w_1 = w_2 = \dots = w_{2d}, \quad w_{2d+1} = w_{2d+2} = \dots = w_{2d^2}, \\ w_0 = 1 - 2dw_1 - 2d(d-1)w_{2d+1}. \quad (9)$$

In addition, the zeroth- to second-order moments of the equilibrium distribution function f_i^{eq} in Eq. (8) should satisfy the following conditions [12],

$$\sum_i f_i^{eq}(\mathbf{x}, t) = \sum_i f_i(\mathbf{x}, t) = \theta(\mathbf{x}, t), \quad (10a)$$

¹ The notation $A_k B_k$ represents the fact that the repeated indexes are summed, i.e., the Einstein notation: $A_k B_k = \sum_k A_k B_k$.

$$\sum_i \mathbf{c}_{i\alpha} f_i^{eq}(\mathbf{x}, t) = 0, \quad (10b)$$

$$\sum_i \mathbf{c}_{i\alpha} \mathbf{c}_{i\beta} f_i^{eq}(\mathbf{x}, t) = c_s^2 \theta(\mathbf{x}, t) \delta_{\alpha\beta}, \quad (10c)$$

where c_s is the lattice sound speed. From Eq. (9), one can obtain $c_s^2 = (2w_1 + 4(d - 1)w_{1+2d})c^2$.

3.2 The Macroscopic Finite-Difference Scheme and the Modified Equation of the MRT-LB Model

To analyze the consistency of the MRT-LB model (6) with the coupled Burgers' equations (1), in this part, we will first present the macroscopic finite-difference scheme of the developed MRT-LB model (6), and then derive the modified equation at fourth order of the macroscopic finite-difference scheme through the Maxwell iteration method.

According to the previous work [3], one can obtain the following macroscopic finite-difference scheme of the MRT-LB model (6) for the d -dimensional diffusion equation (4),

$$\det(T_{\Delta t}^1 \mathbf{I} - \mathbf{A}) \mathbf{m} = \text{adj}(T_{\Delta t}^1 \mathbf{I} - \mathbf{A}) \mathbf{B} \mathbf{m}^{eq}, \quad (11)$$

where $\det(\cdot)$ and $\text{adj}(\cdot)$ denote the determinant and adjoint matrix, respectively. The matrices $\mathbf{A} = \mathbf{W}(\mathbf{I} - \mathbf{S})$, $\mathbf{B} = \mathbf{W}\mathbf{S}$ and the moments $\mathbf{m} = \mathbf{M}\mathbf{f}$, $\mathbf{m}^{eq} = \mathbf{M}\mathbf{f}^{eq}$ with

$$\begin{aligned} \mathbf{W} &= \mathbf{M}\mathbf{T}\mathbf{M}^{-1}, \quad \mathbf{T} = \text{diag}(T_{\Delta x}^{-\mathbf{e}_1}, T_{\Delta x}^{-\mathbf{e}_2}, \dots, T_{\Delta x}^{-\mathbf{e}_q}), \\ \mathbf{f} &= (f_1, f_2, \dots, f_q), \quad \mathbf{f}^{eq} = (f_1^{eq}, f_2^{eq}, \dots, f_q^{eq}), \end{aligned}$$

where $\mathbf{e}_i = \mathbf{c}_i/c$. The time shift operator $T_{\Delta t}^1$ and the space shift operator $T_{\Delta x}^{-\mathbf{e}_k}$ are defined as [3, 5]

$$\begin{aligned} T_{\Delta t}^1(h(\mathbf{x}, t)) &:= h(\mathbf{x}, t + \Delta t), \\ T_{\Delta x}^{-\mathbf{e}_k}(h(\mathbf{x}, t)) &:= h(\mathbf{x} - \mathbf{e}_k \Delta x, t) = h(\mathbf{x} - \mathbf{c}_k \Delta t, t), \end{aligned}$$

here $h(\mathbf{x}, t)$ represents an arbitrary function dependent on the space $\mathbf{x}$ and time t .

Then due to the equivalence between the MRT-LB model (6) and the macroscopic finite-difference scheme (11), and based on the fact that for the LB method, at any order, the macroscopic modified equations are equivalent to the Maxwell iteration equations [3, 18], here we only adopt the Maxwell iteration method to derive the fourth-order modified equation of the macroscopic finite-difference scheme (11) rather than the second-order modified equation given by some commonly used asymptotic analysis methods in Ref. [12]. Specifically, we first perform the Maxwell iteration method on the macroscopic finite-difference scheme (11), and obtain

$$\mathbf{m} - \left(\sum_{k=0}^{+\infty} \mathbf{\Gamma}^k \right) \mathbf{m}^{eq} = \mathbf{0}, \quad (12)$$

with

$$\mathbf{\Gamma} = -\mathbf{S}^{-1} \left(T_{\Delta t}^1 \mathbf{W}^{-1} - \mathbf{I} \right). \quad (13)$$

For the time shift operator $T_{\Delta t}^1$ and the matrix $\mathbf{W}^{-1}$ in Eq. (13), we can rewrite them as the following two series expansion expressions,

$$T_{\Delta t}^1 = \exp(\eta \Delta x^2 \partial_t) = \sum_{k=0}^{+\infty} \frac{\eta^k \Delta x^{2k} \partial_t^k}{k!} = 1 + \eta \Delta x^2 \partial_t + \frac{\eta^2 \Delta x^4}{2} \partial_t^2 + O(\Delta x^6), \quad (14a)$$

$$\begin{aligned} \mathbf{W}^{-1} &= \mathbf{M} \mathbf{T}^{-1} \mathbf{M}^{-1} = \mathbf{M} \text{diag} \left(\exp(-\Delta x \mathbf{e}_1 \cdot \nabla), \exp(-\Delta x \mathbf{e}_2 \cdot \nabla), \dots, \exp(-\Delta x \mathbf{e}_q \cdot \nabla) \right) \mathbf{M}^{-1} \\ &:= \exp(\Delta x \mathcal{W}) = \sum_{k=0}^{+\infty} \frac{\Delta x^k \mathcal{W}^k}{k!} \\ &= \mathbf{I} + \Delta x \mathcal{W} + \frac{\Delta x^2}{2} \mathcal{W}^2 + \frac{\Delta x^3}{6} \mathcal{W}^3 + \frac{\Delta x^4}{24} \mathcal{W}^4 + O(\Delta x^5), \end{aligned} \quad (14b)$$

with $\mathcal{W} = \mathbf{M} \text{diag}(\mathbf{e}_1 \cdot \nabla, \mathbf{e}_2 \cdot \nabla, \dots, \mathbf{e}_q \cdot \nabla) \mathbf{M}^{-1}$, where the diffusive scaling has been used, i.e., $\Delta t = \eta \Delta x^2$ ($\eta \in \mathbb{R}^+$), and in particular, η is kept fixed as $\Delta t \rightarrow 0$ and $\Delta x \rightarrow 0$. In addition, with the aid of Eq. (14), we also have

$$\begin{aligned} \Gamma^1 &= -\mathbf{S}^{-1} \left(\Delta x \mathcal{W} + \frac{\Delta x^2}{2} \mathcal{W}^2 + \frac{\Delta x^3}{6} \mathcal{W}^3 + \frac{\Delta x^4}{24} \mathcal{W}^4 + \eta \Delta x^2 \partial_t \left(\Delta x \mathcal{W} + \frac{\Delta x^2}{2} \mathcal{W}^2 \right) \right. \\ &\quad \left. + \frac{\eta \Delta x^4 \partial_t^2}{2} \mathbf{I} \right) + O(\Delta x^5), \end{aligned} \quad (15a)$$

$$\begin{aligned} \Gamma^2 &= \left(-\mathbf{S}^{-1} \left(\Delta x \mathcal{W} + \frac{\Delta x^2}{2} \mathcal{W}^2 + \frac{\Delta x^3}{6} \mathcal{W}^3 + \eta \Delta x^2 \partial_t (\mathbf{I} + \Delta x \mathcal{W}) \right) \right)^2 + O(\Delta x^5) \\ &= \Delta x^2 \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} + \Delta x^3 \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \left(\frac{\mathcal{W}^2}{2} + \eta \partial_t \mathbf{I} \right) + \Delta x^3 \mathbf{S}^{-1} \left(\frac{\mathcal{W}^2}{2} + \eta \partial_t \mathbf{I} \right) \mathbf{S}^{-1} \mathcal{W} \\ &\quad + \Delta x^4 \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \left(\frac{\mathcal{W}^3}{6} + \eta \mathcal{W} \partial_t \right) + \Delta x^4 \mathbf{S}^{-1} \left(\frac{\mathcal{W}^3}{6} + \eta \mathcal{W} \partial_t \right) \mathbf{S}^{-1} \left(\frac{\mathcal{W}^3}{6} + \eta \mathcal{W} \partial_t \right) \\ &\quad + \Delta x^4 \mathbf{S}^{-1} \left(\frac{\mathcal{W}^3}{6} + \eta \mathcal{W} \partial_t \right) \mathbf{S}^{-1} \mathcal{W} + O(\Delta x^5), \end{aligned} \quad (15b)$$

$$\begin{aligned} \Gamma^3 &= \left(-\mathbf{S}^{-1} \left(\Delta x \mathcal{W} + \frac{\Delta x^2}{2} \mathcal{W}^2 + \eta \Delta x^2 \partial_t \mathbf{I} \right) \right)^3 + O(\Delta x^5) = -\Delta x^3 \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} \\ &\quad - \Delta x^4 \mathbf{S}^{-1} \mathcal{W} \left(\left(\mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \left(\frac{\mathcal{W}^2}{2} + \eta \partial_t \mathbf{I} \right) \right) + \left(\frac{\mathcal{W}^2}{2} + \eta \partial_t \mathbf{I} \right) \mathbf{S}^{-1} \mathcal{W} \right) + O(\Delta x^5), \end{aligned} \quad (15c)$$

$$\Gamma^4 = \Delta x^4 \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} \mathbf{S}^{-1} \mathcal{W} + O(\Delta x^5). \quad (15d)$$

Here it should be noted that we only expand the time shift operator $T_{\Delta t}^1$, $\mathbf{W}^{-1}$, and Γ^k ($k = 1, 2, 3, 4$) up to $O(\Delta x^5)$, which is sufficient to derive the fourth-order modified equation of the macroscopic finite-difference scheme (11), i.e., the following truncation equations up to $O(\Delta x^5)$ of Eq. (12),

$$\mathbf{m} - \mathbf{m}^{eq} - \left(\sum_{k=1}^4 \Gamma^k \right) \mathbf{m}^{eq} = O(\Delta x^5). \quad (16)$$

Substituting Eq. (15) into the above Eq. (16), and with some algebraic and symbolic manipulations, it is easy to calculate the moments $\mathbf{m}^{eq}$ and $\mathbf{m}$, $\mathcal{W}$, and Γ^k ($k = 1, 2, 3, 4$) from the equilibrium distribution function $\mathbf{f}^{eq}$, moment conditions, lattice structure, transformation

matrix $\mathbf{M}$, and relaxation matrix $\mathbf{S}$ of the MRT-LB model (6). Then selecting the first row of Eq. (16) corresponding to the conservative moment θ yields the following fourth-order modified equations,

$d = 1$:

$$\frac{\partial \theta}{\partial t} = 2w_1 \left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{1}{\eta} \frac{\partial^2 \theta}{\partial x^2} + \Delta x^2 \frac{w_1(s_1 - 2)R_1}{12s_1^3 s_2} \frac{\partial^4 \theta}{\partial x^4} + O(\Delta x^4), \quad (17)$$

with

$$R_1 = 6(s_2(s_1 - 1) + s_1^2 + s_2 - 2s_1) - s_2 s_1^2 + 6w_1(2s_1(2 - s_1) + s_2(4 + s_1^2 - 6s_1)), \quad (18)$$

where

$$\begin{aligned} \frac{\partial^2 \theta}{\partial t^2} &= \left(2w_1 \left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{\Delta x^2}{\Delta t} \right)^2 \frac{\partial^4 \theta}{\partial x^4} + O(\Delta x^2), \\ \frac{\partial^3 \theta}{\partial x^2 \partial t} &= \left(2w_1 \left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{\Delta x^2}{\Delta t} \right) \frac{\partial^4 \theta}{\partial x^4} + O(\Delta x^2), \end{aligned}$$

have been used to get rid of high-order time derivatives and spatio-temporal derivatives in the modified equations [15, 16, 42, 61].

$d > 1$:

$$\begin{aligned} \frac{\partial \theta}{\partial t} &= \left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{(2w_1 + 4(d-1)w_{1+2d})}{\eta} \left(\frac{\partial^2 \theta}{\partial x_1^2} + \frac{\partial^2 \theta}{\partial x_2^2} + \dots + \frac{\partial^2 \theta}{\partial x_d^2} \right) \\ &\quad + \Delta x^2 \frac{(w_1 + 2(d-1)w_{1+2d})(s_1 - 2)R_{d1}}{12s_1^3 s_{21}} \left(\frac{\partial^4 \theta}{\partial x_1^4} + \frac{\partial^4 \theta}{\partial x_2^4} + \dots + \frac{\partial^4 \theta}{\partial x_d^4} \right) \\ &\quad - \Delta x^2 \frac{R_{d2}}{s_1^3 s_{21} s_{22}} \left(\frac{\partial^4 \theta}{\partial x_1^2 \partial x_2^2} + \frac{\partial^4 \theta}{\partial x_1^2 \partial x_3^2} + \dots + \frac{\partial^4 \theta}{\partial x_1^2 \partial x_d^2} + \dots + \frac{\partial^4 \theta}{\partial x_{d-1}^2 \partial x_d^2} \right) + O(\Delta x^4), \end{aligned} \quad (19)$$

with

$$\begin{aligned} R_{d1} &= 6(s_{21}(s_1 - 1) + s_1^2 + s_{21} - 2s_1) - s_{21}s_1^2 \\ &\quad + 6(w_1 + 2(d-1)w_{1+2d})(2s_1(2 - s_1) + s_{21}(4 + s_1^2 - 6s_1)), \end{aligned} \quad (20a)$$

$$\begin{aligned} R_{d2} &= s_1^2 w_{1+2d} (2(s_1 - 2)(2s_{21} + s_{22}) + s_{21}s_{22}(8 - 3s_1)) - 8s_1 s_{21} s_{22} (w_1 + 2(d-1)w_{1+2d})^2 \\ &\quad + s_{22} (w_1^2 + 4(d-1)w_{1+2d}(w_1 + (d-1)w_{1+2d})) \\ &\quad \times (8(s_1 + s_{21} - s_1^2 + (s_1 - 1)s_1 s_{21}) + (2 - s_{21})s_1^3), \end{aligned} \quad (20b)$$

where

$$\begin{aligned} \frac{\partial^2 \theta}{\partial t^2} &= \left(\left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{(2w_1 + 4(d-1)w_{1+2d})}{\eta} \right)^2 \left(\frac{\partial^4 \theta}{\partial x_1^4} + \frac{\partial^4 \theta}{\partial x_2^4} + \dots + \frac{\partial^4 \theta}{\partial x_d^4} \right) + O(\Delta x^2), \\ \frac{\partial^3 \theta}{\partial x_i^2 \partial t} &= \left(\frac{1}{s_1} - \frac{1}{2} \right) \frac{(2w_1 + 4(d-1)w_{1+2d})}{\eta} \left(\frac{\partial^4 \theta}{\partial x_i^2 \partial x_1^2} + \frac{\partial^4 \theta}{\partial x_i^2 \partial x_2^2} + \dots + \frac{\partial^4 \theta}{\partial x_i^2 \partial x_{i-1}^2} \right. \\ &\quad \left. + \frac{\partial^4 \theta}{\partial x_i^4} + \frac{\partial^4 \theta}{\partial x_i^2 \partial x_{i+1}^2} + \dots + \frac{\partial^4 \theta}{\partial x_i^2 \partial x_d^2} \right) + O(\Delta x^2), \quad i = 1, 2, \dots, d, \end{aligned}$$

have been used.

We now give a remark on Eqs. (17), (18), (19), and (20).

Remark 1 The modified equation at fourth order of the MRT-LB model (or the fourth-order MRT-LB model) for the d -dimensional diffusion equation can be obtained once the free parameters (the weight coefficients and the relaxation parameters) in the MRT-LB model (6) satisfy the fourth-order conditions, i.e.,

$$\begin{aligned} d = 1 : R_1 &= 0, \\ d > 1 : R_{d1} &= 0, \quad R_{d2} = 0, \end{aligned}$$

and the corresponding solution is given by

$$d = 1 : \begin{cases} w_1 = \frac{\varepsilon s_1}{2 - s_1}, \\ s_2 = \frac{6(s_1 - 2)(s_1 + 2\varepsilon s_1 - 2)}{24\varepsilon - 8s_1 - 36\varepsilon s_1 + 6\varepsilon s_1^2 + s_1^2 + 12}, \end{cases} \quad (21a)$$

$$d > 1 : \begin{cases} w_1 = \frac{\varepsilon s_1}{2 - s_1} - 2(d - 1)w_5, \\ s_{21} = \frac{6(s_1 - 2)(s_1 + 2\varepsilon s_1 - 2)}{24\varepsilon - 8s_1 - 36\varepsilon s_1 + 6\varepsilon s_1^2 + s_1^2 + 12}, \\ s_{22} = \frac{6w_1 + 2d(s_1 - 2)(s_1 + 2\varepsilon s_1 - 2)}{\varepsilon^2 s_1^2 - 6\varepsilon^2 s_1 + 6\varepsilon^2 + 6w_1 + 2d\varepsilon s_1^2 - 6w_1 + 2d\varepsilon s_1 - 12(d - 1)w_1 + 2d\varepsilon + 4w_1 + 2d s_1^2 - 17w_1 + 2d s_1 + 18(d - 1)w_1 + 2d}. \end{cases} \quad (21b)$$

From Eq. (21), it can be clearly seen that the relaxation parameter s_1 and the weight coefficient w_{1+2d} in the MRT-LB model (6) still remain free. These parameters can be utilized to improve the computational accuracy of the gradient term $\nabla\theta$ (see following Part 3.3 for details).

3.3 The Calculation of the Velocity $\mathbf{u}$ in the Coupled Burgers' Equation

The aim of this work is to obtain a fourth-order MRT-LB model for the d -dimensional coupled Burgers' equations (1), this means that it is necessary to develop a fourth-order scheme to determine the velocity $\mathbf{u}$ with a fourth-order accuracy in space from the variable θ in the diffusion equation (4). To this end, one can adopt the fourth-order finite-difference method to derive the velocity $\mathbf{u}$ from θ , however, this would be more expensive. Here, based on the relation between $\mathbf{u}$ and θ given by Eq. (3), we can utilize the information encoded in the non-conserved moment of the MRT-LB model to obtain a high-order local scheme for the gradient term $\nabla\theta$.

According to the Maxwell iteration method [12], we can expand the distribution function $\mathbf{f}$ at the equilibrium distribution function $\mathbf{f}^{eq}$ as

$$\begin{aligned} \mathbf{f} = & \mathbf{f}^{eq} - \Delta x \Lambda^{-1} \bar{\mathbf{D}} \mathbf{f}^{eq} - \eta \Delta x^2 \Lambda^{-1} \partial_t \mathbf{f}^{eq} - \Delta x^3 \Lambda^{-1} \bar{\mathbf{D}} \left(\frac{\Lambda^{-1} \bar{\mathbf{D}}^2}{2} + \frac{\bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}}}{2} - \frac{\bar{\mathbf{D}}^2}{6} - \Lambda^{-1} \bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}} \right) \mathbf{f}^{eq} \\ & + \eta^2 \Delta x^4 \Lambda^{-1} \bar{\mathbf{D}} \left(\Lambda^{-1} - \frac{\mathbf{I}}{2} \right) \mathbf{D} \mathbf{f}^{eq} \\ & + \Delta x^4 \Lambda^{-1} \bar{\mathbf{D}} \left(-\frac{\bar{\mathbf{D}}^3}{24} + \frac{\Lambda^{-1} \bar{\mathbf{D}}^3}{6} + \frac{\bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}}^2}{4} + \frac{\bar{\mathbf{D}}^2 \Lambda^{-1} \bar{\mathbf{D}}}{6} + \frac{\Lambda^{-1} \bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}}^2}{2} \right. \\ & \quad \left. + \frac{\Lambda^{-1} \bar{\mathbf{D}}^2 \Lambda^{-1} \bar{\mathbf{D}}}{2} + \frac{\bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}}}{2} + \Lambda^{-1} \bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}} \Lambda^{-1} \bar{\mathbf{D}} \right) \mathbf{f}^{eq} + O(\Delta x^5), \end{aligned} \quad (22)$$

with $\bar{\mathbf{D}} = \text{diag}(\mathbf{e}_1 \cdot \nabla, \mathbf{e}_2 \cdot \nabla, \dots, \mathbf{e}_q \cdot \nabla)$ and $\mathbf{D} = \partial_t \mathbf{I}_q + \bar{\mathbf{D}}$. Here we note that there still are some free parameters in the MRT-LB model (6) after obtaining the modified equation at fourth order of the transformed diffusion equation (4) (see Remark 1 for details). Actually, from Eq. (22), some mixed partial terms, where the orders of them are equal to or greater than 2, can be eliminated with the aid of the free parameters. To see this more clearly, we now give a theorem.

Theorem 1 For the d -dimensional coupled Burgers' equations (1), once the following conditions are satisfied,

$$\begin{aligned} d = 1 : \text{Con}_1 &= 3s_1(s_2 + s_1 - 2) - s_1^2 s_2 + 6w_1(2 - s_1)(s_1 + s_2(1 - s_1)) = 0, \\ d > 1 : \begin{cases} \text{Con}_{d1} = 3s_1(s_{21} + s_1 - 2) - s_1^2 s_{21} \\ \quad + 6(w_1 + 2(d-1)w_{1+2d})(2 - s_1)(s_1 + s_{21}(1 - s_1)) = 0, \\ \text{Con}_{d2} = -(s_{22}w_1^2 + 4s_{22}(w_1 + (d-1)w_{1+2d})w_{1+2d}(d-1)) \\ \quad (s_1 - 2)(s_1 + s_{21} - s_1 s_{21}) - s_1^2 w_{1+2d}(2s_{21} + s_{22} - 2s_{21}s_{22}) = 0, \end{cases} \end{aligned}$$

the distribution function $\mathbf{f}$ in Eq. (22) at the diffusive scaling can be simplified as

$$\mathbf{f} = \mathbf{f}^{eq} - \Delta x \mathbf{\Lambda}^{-1} \bar{\mathbf{D}} \mathbf{f}^{eq} + O(\Delta x^5). \quad (23)$$

In addition, from Eq. (23) and the moment conditions in Eq. (10), we can obtain

$$\begin{aligned} \sum_{k=1}^q \mathbf{c}_k (f_k - f_k^{eq}) &= \sum_{k=1}^q \mathbf{c}_k f_k^{ne} = \sum_{k=1}^q \mathbf{c}_k f_k = -\frac{1}{s_1} \Delta t \left(\nabla \cdot \sum_{k=1}^q \mathbf{c}_k \mathbf{c}_k f_k^{eq} \right) + O(\Delta x^4) \\ &= -\frac{(2w_1 + 4(d-1)w_{1+2d})}{s_1 \eta} \nabla \theta + O(\Delta x^4), \end{aligned}$$

which means that the gradient term $\nabla \theta$ can be determined locally by

$$\nabla \theta = -\frac{s_1 \eta}{2w_1 + 4(d-1)w_{1+2d}} \sum_{k=1}^q \mathbf{c}_k f_k^{ne}, \quad (24)$$

which is fourth-order accurate in space. Then based on Eqs. (3) and (10), the velocity $\mathbf{u}$ in the d -dimensional coupled Burgers' equations (1) can be calculated with the fourth-order accuracy by the following formula,

$$u_i(\mathbf{x}, t) = -2\nu \frac{\partial_{x_i} \theta(\mathbf{x}, t)}{\theta(\mathbf{x}, t)} = (2 - s_1) \frac{\sum_{k=1}^q \mathbf{c}_{ki} f_k^{ne}(\mathbf{x}, t)}{\sum_{k=1}^q f_k(\mathbf{x}, t)}, \quad i = 1, 2, \dots, d. \quad (25)$$

3.4 The Conditions of the Fourth-Order MRT-LB Model

Now we give a theorem to present the conditions to ensure that the MRT-LB model for the d -dimensional coupled Burgers' equations, i.e., Eqs. (6) and (25), has a fourth-order accuracy in space.

Theorem 2 The Cole–Hopf transformation based MRT-LB model (see Eqs. (6) and (25)) for the d -dimensional coupled Burgers' equations (1) is fourth-order accurate once the weight coefficients and the relaxation parameters are given by

$$\begin{aligned} d = 1 : \begin{cases} \varepsilon = 2w_1 \xi \quad (2\text{nd order consistency}), \\ R_1 = 0 \quad (4\text{th order consistency}), \\ \text{Con}_1 = 0 \quad (\text{simplified local gradient reconstruction}), \end{cases} \\ d > 1 : \begin{cases} \varepsilon = (2w_1 + 4(d-1)w_{1+2d}) \xi \quad (2\text{nd order consistency}), \\ R_{di} = 0, \quad i = 1, 2 \quad (4\text{th order consistency}), \\ \text{Con}_{di} = 0, \quad i = 1, 2 \quad (\text{simplified local gradient reconstruction}), \end{cases} \end{aligned} \quad (26)$$

where $\xi = 1/s_1 - 1/2$ and $\varepsilon = \nu\eta$. From Eq. (26), one can determine the weight coefficients and relaxation parameters after some algebraic manipulations,

$$d = 1 : \begin{cases} s_1 = 2/(6\varepsilon + 1), \\ s_2 = 24\varepsilon/(6\varepsilon + 1)^2, \\ w_1 = 1/6, \end{cases} \quad d > 1 : \begin{cases} s_1 = 2/(6\varepsilon + 1), \\ s_{21} = 24\varepsilon/(6\varepsilon + 1)^2, \\ s_{22} = 4/(6\varepsilon + 3), \\ w_1 = 1/6 - (d-1)\varepsilon/3, \\ w_{1+2d} = \varepsilon/6. \end{cases} \quad (27)$$

Additionally, the gradient term $\nabla\theta$ in the diffusion equation (4) can also be calculated locally by the non-equilibrium distribution function (see Eq. (24)) with a fourth-order accuracy in space.

For the relaxation parameters determined in Eq. (27), we now give the following remark.

Remark 2 At the mesoscopic level, the LB method based on the evolution of the distribution function is a two-level scheme. However, the previous work [5] demonstrates that the LB model can be rewritten as a macroscopic multiple-level finite-difference scheme (see Eq. (11)) in which the time level depends on the number of the relaxation parameters not equal to 1. Thus, according to the relaxation parameters determined by Eq. (27) in Theorem 2 and the expressions of the relaxation matrix in Eqs. (7a), (7b), (7c), and (30), we can conclude that the macroscopic finite-difference scheme of the developed fourth-order MRT-LB model for the d -dimensional coupled Burgers' equations is a two-level scheme when $\varepsilon = 1/6$, and it is a $(d^2/2 + 3d/2 + 1)$ -level scheme when $\varepsilon \neq 1/6$. This shows the high efficiency of the mesoscopic LB method from a computational point of view.

4 Numerical Results and Discussion

In this section, we will conduct some numerical experiments with the fourth-order MRT-LB model where the weight coefficient and the relaxation parameters are determined from Eqs. (9) and (27). To evaluate the difference between the analytical and numerical results, the root-mean-square error (RMSE) is adopted,

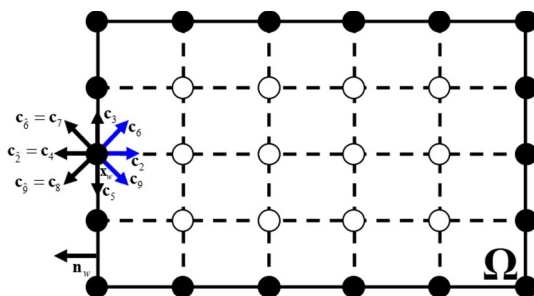
$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^d \sum_{j_i=1}^{N_{x_i}} \left(\psi((j_1, j_2, \dots, j_d)\Delta x, n_t \Delta t) - \psi^*((j_1, j_2, \dots, j_d)\Delta x, n_t \Delta t) \right)^2}{\prod_{i=1}^d N_{x_i}}},$$

where N_{x_i} ($i = 1, 2, \dots, d$) is the number of grid points, $n_t = t/\Delta t$ is the total number of time steps with t being the final time of the numerical simulation, ψ and ψ^* are the numerical and analytical solutions. Then, we can further estimate the convergence rate (CR) of the developed MRT-LB model.

Before performing the numerical simulations, we would like to point out that to preserve the fourth-order accuracy of the present MRT-LB model for the d -dimensional coupled Burgers' equations, the initialization and boundary schemes of the distribution function $f_i(\mathbf{x}, t)$ in the MRT-LB model must be properly given.

- For the initial condition given by Eq. (5a), we can adopt Eq. (23) to approximate the unknown distribution function f_i , which is fifth-order accurate in space and can maintain the fourth-order accuracy of the present MRT-LB model [4, 6, 54].
- For the boundary scheme of the unknown distribution function $f_i(\mathbf{x}_w, t)$ ($\mathbf{x}_w \in \partial\Omega$) during the streaming process of the MRT-LB model, here we only discuss the homogeneous

Fig. 1 The dotted line is the lattice line, and the solid line is the boundary. White and black circles are the lattice nodes in Ω and the intersections of the boundary and the lattice line, respectively



Dirichlet boundary condition in Eq. (2b) for the coupled Burgers' equation (1), i.e., the homogeneous Neumann boundary condition given by Eq. (5b) in the diffusion equation (4). From Eq. (23), we have

$$f_i(\mathbf{x}_w, t) - f_{\hat{i}}(\mathbf{x}_w, t) = f_i^{eq}(\mathbf{x}_w, t) - f_{\hat{i}}^{eq}(\mathbf{x}_w, t) - \Delta x \Lambda_{ik}(\mathbf{e}_k \cdot \nabla f_k^{eq}(\mathbf{x}_w, t)) + \Delta t \Lambda_{ik}(\mathbf{e}_k \cdot \nabla f_k^{eq}(\mathbf{x}_w, t)) + O(\Delta x^5), \quad (28)$$

where the index $\hat{i}$ satisfies $\mathbf{c}_{\hat{i}} = \mathbf{c}_i - 2c\mathbf{n}_w$ (see the relation between $\mathbf{c}_i$ and $\mathbf{c}_{\hat{i}}$ of the two-dimensional case in Fig. 1).

Then based on Eq. (5b), we can simplify Eq. (28) as (the derivation can be found in Appendix B)

$$f_i(\mathbf{x}_w, t) = f_{\hat{i}}(\mathbf{x}_w, t) + O(\Delta x^5). \quad (29)$$

From the above equation, it is obvious that the boundary scheme of the distribution function $f_i(\mathbf{x}_w, t)$ for the homogeneous Neumann boundary condition in Eq. (5b) is fifth-order accurate.

Example 1 We first consider the one-dimensional Burgers' equation (1) with the following initial and boundary conditions,

$$\begin{cases} u(x, 0) = \sin(\pi x), & 0 \leq x \leq 1, \\ u(0, t) = u(1, t) = 0, & t > 0, \end{cases}$$

and the analytical solution of u is given by

$$u(x, t) = 4\pi v \frac{\sum_{n=1}^{+\infty} a_n \exp(-n^2 \pi^2 v t) n \sin(n\pi x)}{a_0 + 2 \sum_{n=1}^{+\infty} a_n \exp(-n^2 \pi^2 v t) \cos(n\pi x)},$$

where $a_n = I_n(1/(2\pi v))$ with $I_n(x)$ representing the first type of the n -th modified Bessel function. With the help of the Cole–Hopf transformation, one can derive the analytical solution of θ in the one-dimensional diffusion equation,

$$\theta(x, t) = a_0 + 2 \sum_{n=1}^{+\infty} a_n \exp(-n^2 \pi^2 v t) \cos(n\pi x),$$

with the initial and boundary conditions

$$\begin{cases} \theta(x, 0) = a_0 + 2 \sum_{n=1}^{+\infty} a_n \cos(n\pi x), & 0 \leq x \leq 1, \\ \partial_x \theta(0, t) = \partial_x \theta(1, t) = 0, & t > 0. \end{cases}$$

In the following simulations, the maximal value of n is taken as 100000.

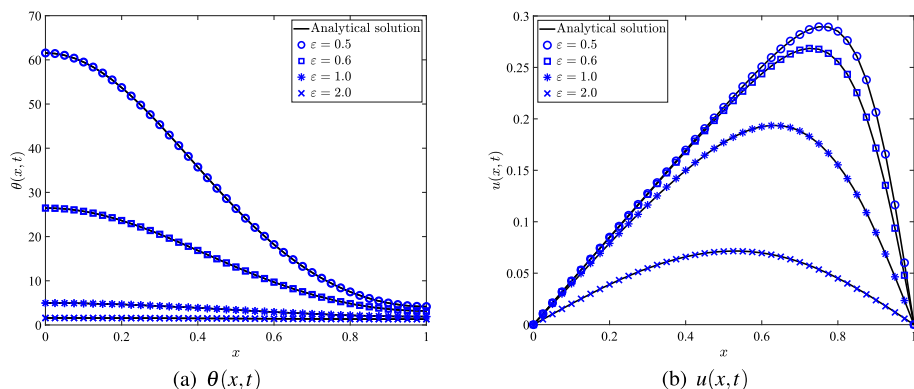


Fig. 2 Comparisons of the analytical and numerical results under different values of the parameter ε ($t = 2.0$) (solid lines: analytical solutions, symbols: numerical results)

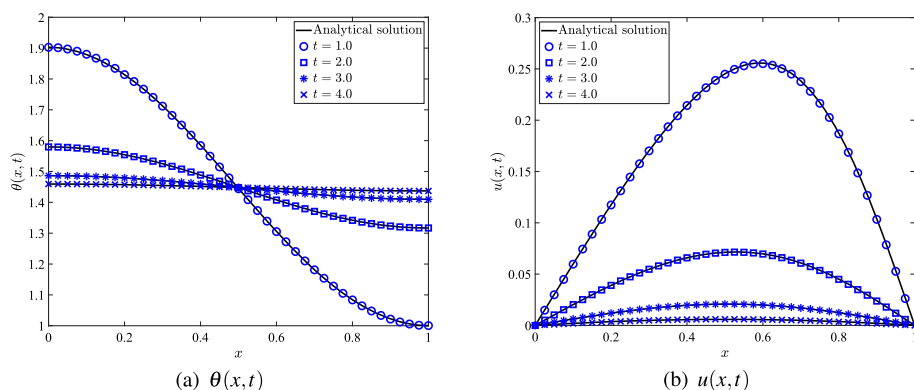


Fig. 3 Comparisons of the analytical and numerical results at different values of the time t ($\varepsilon = 2.0$) (solid lines: analytical solutions, symbols: numerical results)

We first conduct some simulations under different values of the parameter ε ($\varepsilon = 0.5, 0.6, 1.0, 2.0$) corresponding to different viscosity coefficients for the specific lattice spacing $\Delta x = 1/40$ and time step $\Delta t = 1/100$, and plot the results in Fig. 2. Additionally, to see the evolutions of variables θ and u in time, we also carry out some simulations at different values of the time t ($t = 1.0, 2.0, 3.0, 4.0$), and present the results in Fig. 3 where $\varepsilon = 2.0$. As shown in Figs. 2 and 3, the numerical results of the MRT-LB model are in good agreement with the corresponding analytical solutions.

In order to measure the deviation between the numerical results and analytical solutions, we also perform some simulations under different values of the lattice spacing Δx , and list the RMSEs and average CRs (CRs)² of the MRT-LB model in Table 1. From the result shown in this table, one can observe that the MRT-LB model is fourth-order accurate in space, which is in agreement with our theoretical analysis.

² The average CR is defined as $\overline{\text{CR}} = \left(\sum_{i=1}^L \log \left(\text{RMSE}_{\Delta x/2^{i-1}, \Delta t/4^{i-1}} / \text{RMSE}_{\Delta x/2^i, \Delta t/4^i} \right) \right) / (L \log 2)$.

Table 1 The RMSEs and $\overline{\text{CR}}$ s under different values of the parameter ε ($t = 2.0$)

ε	Δx	Δt	variable	$\text{RMSE}_{\Delta x, \Delta t}$	$\text{RMSE}_{\Delta x}/2, \Delta t/4$	$\text{RMSE}_{\Delta x}/4, \Delta t/16$	$\text{RMSE}_{\Delta x}/8, \Delta t/64$	$\overline{\text{CR}}$
0.5	$\frac{1}{40}$	$\frac{1}{100}$	θ	7.8613×10^{-4}	4.8926×10^{-5}	3.0595×10^{-6}	1.9139×10^{-7}	4.0013
			u	1.9827×10^{-5}	1.1821×10^{-6}	7.2429×10^{-8}	4.4865×10^{-9}	4.0365
1	$\frac{1}{40}$	$\frac{1}{100}$	θ	2.4232×10^{-4}	1.5014×10^{-5}	9.3770×10^{-7}	5.8641×10^{-8}	4.0042
			u	3.2184×10^{-5}	2.0194×10^{-6}	1.2608×10^{-7}	7.8713×10^{-9}	3.9992
1.5	$\frac{1}{40}$	$\frac{1}{100}$	θ	2.3106×10^{-4}	1.3918×10^{-5}	8.6303×10^{-7}	5.3874×10^{-8}	4.0221
			u	1.0796×10^{-4}	6.4468×10^{-6}	3.9752×10^{-7}	2.4739×10^{-8}	4.0305
2	$\frac{1}{40}$	$\frac{1}{100}$	θ	2.1330×10^{-4}	1.2349×10^{-5}	7.5838×10^{-7}	4.7229×10^{-8}	4.0470
			u	1.8574×10^{-4}	1.0638×10^{-5}	6.4967×10^{-7}	4.0343×10^{-8}	4.0562

Example 2. We consider the **two-dimensional coupled Burgers' equations** (1) with periodic boundary conditions and the following initial conditions,

$$\begin{cases} u_x(x, y, 0) = -\frac{4\nu\pi \cos(2\pi x) \sin(\pi y)}{2 + \sin(2\pi x) \sin(\pi y)}, & (x, y) \in \Omega, \\ u_y(x, y, 0) = -\frac{2\nu\pi \sin(2\pi x) \cos(\pi y)}{2 + \sin(2\pi x) \sin(\pi y)}, & (x, y) \in \Omega, \end{cases}$$

where the computational domain $\Omega = \{(x, y) : 0 \leq x, y \leq 2\}$. Under the condition of Eq. (??), one can obtain the analytical solution of $\mathbf{u} = (u_x, u_y)$,

$$\begin{cases} u_x(x, y, t) = -\frac{4\nu\pi \exp(-5\nu\pi^2 t) \cos(2\pi x) \sin(\pi y)}{2 + \exp(-5\nu\pi^2 t) \sin(2\pi x) \sin(\pi y)}, \\ u_y(x, y, t) = -\frac{2\nu\pi \exp(-5\nu\pi^2 t) \sin(2\pi x) \cos(\pi y)}{2 + \exp(-5\nu\pi^2 t) \sin(2\pi x) \sin(\pi y)}, \end{cases}$$

then according to Eq. (4), the analytical solution of θ can be derived [65],

$$\theta(x, y, t) = \frac{2 + \exp(-5\nu\pi^2 t) \sin(2\pi x) \sin(\pi y)}{2}.$$

First of all, for the given lattice spacing $\Delta x = 1/40$ and time step $\Delta t = 1/100$, we set the parameter $\varepsilon = 0.2$ to determine the viscosity coefficient ν , and measure the absolute errors between the analytical and numerical results ($|\psi^* - \psi|$, $\psi = \theta, u_x$, and u_y) of the variables θ, u_x , and u_y in Fig. 4 where $t = 2.0$. As shown in this figure, the maximum absolute errors of the variables θ, u_x , and u_y are less than 2.5×10^{-7} , 1.2×10^{-7} , and 7.0×10^{-9} , respectively. Then we also conduct some simulations at different values of the time t ($t = 2.0, 3.0, 4.0, 5.0$), and plot the profiles of the variables θ, u_x , and u_y in Fig. 5 where $\varepsilon = 0.2$. As seen from this figure, the numerical results are very close to the analytical solutions. Finally, to test the CR of the MRT-LB model, besides above periodic boundary conditions, we also consider the following homogeneous Neumann boundary conditions,

$$\begin{cases} \partial_x \theta(1/4, y, t) = \partial_x \theta(9/4, y, t) = 0, & 1/2 \leq y \leq 5/2, \\ \partial_y \theta(x, 1/2, t) = \partial_y \theta(x, 5/2, t) = 0, & 1/4 \leq x \leq 9/4, \end{cases}$$

where the computational domain is given by $\Omega = \{(x, y) : 1/4 \leq x \leq 9/4, 1/2 < y < 5/2\}$. We measure the RMSEs between the numerical and analytical solutions under different values of the parameter ε , and calculate the CRs in Tables 2 and 3 where $t = 2.0$. From these tables, one can observe that the for the two-dimensional coupled Burgers' equations with either periodic boundary conditions or homogeneous Dirichlet boundary conditions, the developed MRT-LB model have a fourth-order convergence rate in space.

Example 2 We now focus on the **three-dimensional coupled Burgers' equations** (1) with periodic boundary conditions and the following initial conditions,

$$\begin{cases} u_x(x, y, z, 0) = -\frac{4\nu\pi \cos(2\pi x) \sin(\pi y) \sin(4\pi z)}{2 + \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, & (x, y, z) \in \Omega, \\ u_y(x, y, z, 0) = -\frac{2\nu\pi \sin(2\pi x) \cos(\pi y) \sin(4\pi z)}{2 + \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, & (x, y, z) \in \Omega, \\ u_z(x, y, z, 0) = -\frac{8\nu\pi \sin(2\pi x) \sin(\pi y) \cos(4\pi z)}{2 + \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, & (x, y, z) \in \Omega, \end{cases}$$

Table 2 The RMSEs and $\overline{CR}$ s under different values of the parameter ε ($t = 2.0$, periodic boundary)

ε	Δx	Δt	variable	$RMSE_{\Delta x, \Delta t}$	$RMSE_{\Delta x/2, \Delta t/4}$	$RMSE_{\Delta x/4, \Delta t/16}$	$RMSE_{\Delta x/8, \Delta t/64}$	$\overline{CR}$
0.1	$\frac{1}{20}$	$\frac{1}{50}$	θ	4.2630×10^{-7}	2.6622×10^{-8}	1.6687×10^{-9}	1.0453×10^{-10}	3.9979
			u_x	2.6683×10^{-7}	1.6587×10^{-8}	1.0353×10^{-9}	6.4689×10^{-11}	4.0034
			u_y	3.8117×10^{-8}	2.3404×10^{-9}	1.4282×10^{-10}	8.9076×10^{-12}	4.0210
0.2	$\frac{1}{20}$	$\frac{1}{50}$	θ	1.5114×10^{-6}	9.4845×10^{-8}	5.9523×10^{-9}	3.7300×10^{-10}	3.9948
			u_x	1.5218×10^{-7}	9.2774×10^{-9}	5.7625×10^{-10}	3.5958×10^{-11}	4.0157
			u_y	1.3577×10^{-7}	8.4954×10^{-9}	5.3110×10^{-10}	3.3196×10^{-11}	3.9993
0.3	$\frac{1}{20}$	$\frac{1}{50}$	θ	5.7600×10^{-6}	3.5539×10^{-7}	2.2209×10^{-8}	1.3902×10^{-9}	4.0055
			u_x	1.3681×10^{-6}	8.4242×10^{-8}	5.2444×10^{-9}	3.2745×10^{-10}	4.0095
			u_y	7.7594×10^{-7}	4.8271×10^{-8}	3.0124×10^{-9}	1.8820×10^{-10}	4.0032
0.4	$\frac{1}{20}$	$\frac{1}{50}$	θ	9.9386×10^{-6}	5.9304×10^{-7}	3.6764×10^{-8}	2.2967×10^{-9}	4.0264
			u_x	4.4801×10^{-6}	2.6578×10^{-7}	1.6396×10^{-8}	1.0214×10^{-9}	4.0329
			u_y	2.1821×10^{-6}	1.3113×10^{-7}	8.1137×10^{-9}	5.0581×10^{-10}	4.0249

Table 3 The RMSEs and $\overline{CR}$ s under different values of the parameter ε ($t = 2.0$, homogeneous Neumann boundary)

ε	Δx	Δt	variable	$RMSE_{\Delta x, \Delta t}$	$RMSE_{\Delta x/2, \Delta t/4}$	$RMSE_{\Delta x/4, \Delta t/16}$	$RMSE_{\Delta x/8, \Delta t/64}$	$\overline{CR}$
0.1	$\frac{1}{20}$	$\frac{1}{50}$	θ	4.4762×10^{-7}	2.7287×10^{-8}	1.6895×10^{-9}	1.0518×10^{-10}	4.0213
			u_x	2.6686×10^{-7}	1.6588×10^{-8}	1.0354×10^{-9}	6.4690×10^{-11}	4.0034
			u_y	3.8145×10^{-8}	2.3052×10^{-9}	1.4285×10^{-10}	8.9084×10^{-12}	4.0184
0.2	$\frac{1}{20}$	$\frac{1}{50}$	θ	1.5870×10^{-6}	9.7216×10^{-8}	6.0267×10^{-9}	3.7533×10^{-10}	4.0153
			u_x	1.5260×10^{-7}	9.2768×10^{-9}	5.7623×10^{-10}	3.5958×10^{-11}	4.0157
			u_y	1.3578×10^{-7}	8.4958×10^{-9}	5.3111×10^{-10}	3.3197×10^{-11}	3.9993
0.3	$\frac{1}{20}$	$\frac{1}{50}$	θ	6.0480×10^{-6}	3.6427×10^{-7}	2.2487×10^{-8}	1.3989×10^{-9}	4.0260
			u_x	1.3691×10^{-6}	8.4242×10^{-8}	5.2444×10^{-9}	3.2745×10^{-10}	4.0095
			u_y	7.7595×10^{-7}	4.8271×10^{-8}	3.0124×10^{-9}	1.8820×10^{-10}	4.0032
0.4	$\frac{1}{20}$	$\frac{1}{50}$	θ	1.0435×10^{-5}	6.0787×10^{-7}	3.7223×10^{-8}	2.3110×10^{-9}	4.0469
			u_x	4.4801×10^{-6}	2.6578×10^{-7}	1.6396×10^{-8}	1.0214×10^{-9}	4.0329
			u_y	2.1821×10^{-6}	1.3114×10^{-7}	8.1137×10^{-9}	5.0582×10^{-10}	4.0249

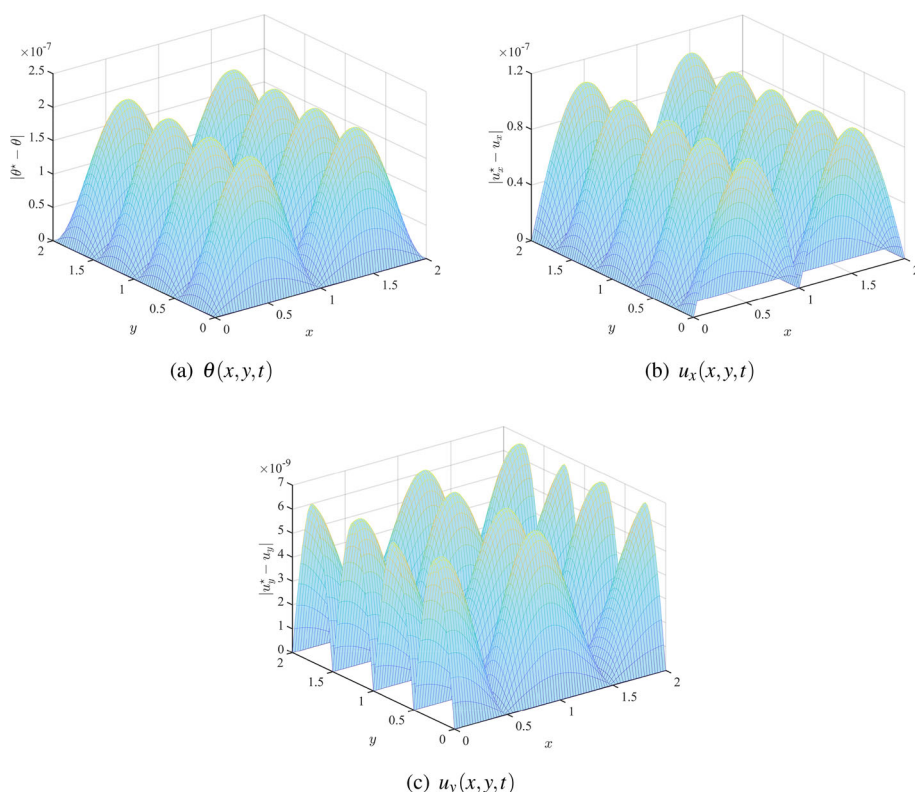


Fig. 4 The absolute errors between the analytical and numerical results of variables θ , u_x , and u_y ($\varepsilon = 0.2$, $t = 2.0$)

where the computational domain is $\Omega = \{(x, y, z) : 0 \leq x, y, z \leq 2\}$. The analytical solution of $\mathbf{u} = (u_x, u_y, u_z)$ is given by

$$\begin{cases} u_x(x, y, z, t) = -\frac{4\nu\pi \exp(-21\nu\pi^2 t) \cos(2\pi x) \sin(\pi y) \sin(4\pi z)}{2 + \exp(-21\nu\pi^2 t) \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, \\ u_y(x, y, z, t) = -\frac{2\nu\pi \exp(-21\nu\pi^2 t) \sin(2\pi x) \cos(\pi y) \sin(4\pi z)}{2 + \exp(-21\nu\pi^2 t) \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, \\ u_z(x, y, z, t) = -\frac{8\nu\pi \exp(-21\nu\pi^2 t) \sin(2\pi x) \sin(\pi y) \cos(4\pi z)}{2 + \exp(-21\nu\pi^2 t) \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}, \end{cases}$$

from which one can also obtain the analytical solution of θ ,

$$\theta(x, y, z, t) = \frac{2 + \exp(-21\nu\pi^2 t) \sin(2\pi x) \sin(\pi y) \sin(4\pi z)}{2}.$$

Similar to the Example 2, we fix the lattice spacing $\Delta x = 1/40$ and time step $\Delta t = 1/100$ in the following simulations, and present the absolute errors between the analytical and numerical solutions of the variables θ , u_x , u_y , and u_z in Fig. 6 where $\varepsilon = 0.2$ and $t = 2.0$. From this figure, one can find that the maximum absolute errors of the variables θ , u_x , u_y ,

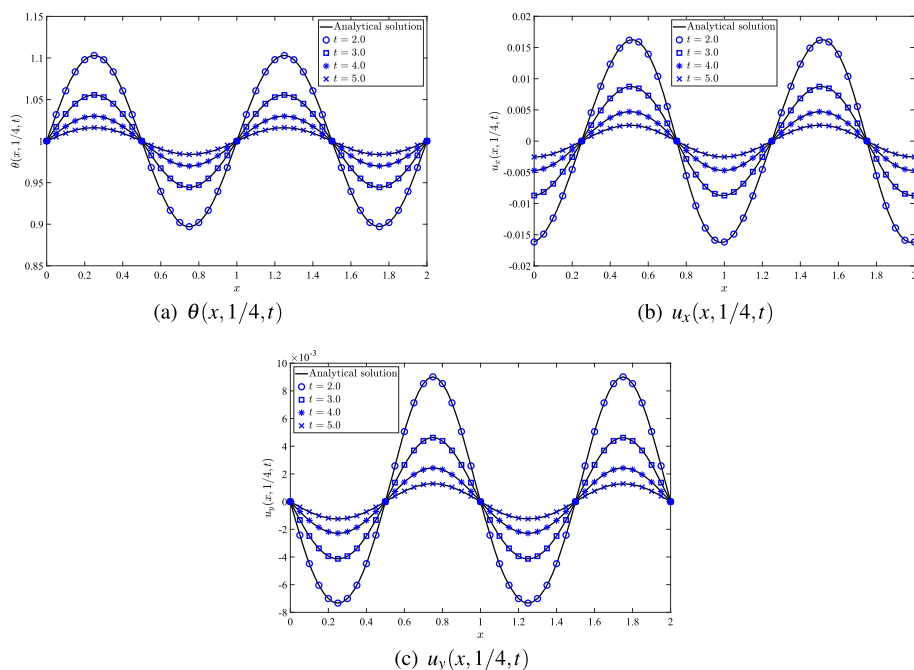


Fig. 5 Profiles of the variables θ , u_x , and u_y at different values of the time t ($\varepsilon = 0.2$) (solid lines: analytical solutions, symbols: numerical results)

and u_z are less than 1.6×10^{-15} , 5×10^{-15} , 3.2×10^{-16} , and 1.2×10^{-7} , respectively. In order to see the evolutions of variables θ , u_x , u_y , and u_z in time more clearly, we further carry out some simulations at different values of the time t ($t = 2.0, 3.0, 4.0, 5.0$) and $\varepsilon = 0.2$, and plot the profiles of the variables θ , u_x , u_y , and u_z in Fig. 7. As shown in this figure, the numerical results are in good agreement with the analytical solutions. Additionally, we also take different values of the parameter ε ($\varepsilon = 0.05, 0.1, 0.15, 0.25$) to calculate the RMSEs and CRs at the time $t = 2.0$, and show the numerical results in Table 4 where a fourth-order accuracy in space is clearly observed. Finally, we also consider the four-dimensional case, and the results are similar to those of the above two- and three-dimensional cases, but for brevity, they are not presented here.

5 Conclusions

In this paper, we first adopted the Cole–Hopf transformation to eliminate the nonlinear convection terms in the $d(\geq 1)$ -dimensional coupled Burgers' equations, and in this case, a simple diffusion equation is obtained. In particular, the velocity $\mathbf{u}$ in the d -dimensional coupled Burgers' equations can be determined by the variable θ and the gradient term $\nabla\theta$ in the transformed diffusion equation. Then at the diffusive scaling, we obtained the macroscopic finite-difference scheme of the MRT-LB model for the d -dimensional transformed diffusion equation, and also the modified equation at fourth order of the MRT-LB model through the

Table 4 The RMSEs and CRs under different values of the parameter ε ($t = 2.0$)

ε	Δx	Δt	variable	$\text{RMSE}_{\Delta x, \Delta t}$	$\text{RMSE}_{\Delta x/2, \Delta t/4}$	$\text{RMSE}_{\Delta x/4, \Delta t/16}$	$\text{RMSE}_{\Delta x/8, \Delta t/64}$	CR
0.05	1/10	1/25	θ	1.4670×10^{-5}	7.9252×10^{-7}	4.7901×10^{-8}	2.9814×10^{-9}	4.0882
			u_x	2.5774×10^{-6}	1.2644×10^{-7}	7.3962×10^{-9}	4.5508×10^{-10}	4.1558
			u_y	1.3175×10^{-6}	6.9426×10^{-8}	4.1467×10^{-9}	2.5654×10^{-10}	4.1088
			u_z	2.1628×10^{-6}	9.9075×10^{-8}	5.5962×10^{-9}	3.4079×10^{-10}	4.2106
0.10	1/10	1/25	θ	1.3769×10^{-7}	7.9947×10^{-9}	4.9629×10^{-10}	3.1115×10^{-11}	4.0372
			u_x	4.5349×10^{-8}	7.9947×10^{-9}	4.9629×10^{-10}	3.1115×10^{-11}	4.0727
			u_y	2.1041×10^{-8}	1.1906×10^{-9}	7.2976×10^{-11}	4.5464×10^{-12}	4.0587
			u_z	5.8170×10^{-8}	3.3611×10^{-9}	2.0675×10^{-10}	1.2891×10^{-11}	4.0466
0.15	1/10	1/25	θ	1.4106×10^{-9}	8.1745×10^{-11}	5.0850×10^{-12}	3.3828×10^{-13}	4.0086
			u_x	6.5890×10^{-10}	3.7145×10^{-11}	2.2804×10^{-12}	1.4216×10^{-13}	4.0594
			u_y	2.5730×10^{-10}	1.4675×10^{-11}	9.0264×10^{-13}	5.6299×10^{-14}	4.0527
			u_z	9.7287×10^{-10}	5.6602×10^{-11}	2.1824×10^{-13}	3.4959×10^{-12}	4.0407
0.20	1/10	1/25	θ	6.4539×10^{-11}	2.9193×10^{-12}	1.7533×10^{-13}	9.5754×10^{-15}	4.2395
			u_x	3.8868×10^{-11}	1.7540×10^{-12}	1.0422×10^{-13}	6.5206×10^{-15}	4.1804
			u_y	1.7096×10^{-11}	7.9028×10^{-13}	4.7037×10^{-14}	3.0432×10^{-15}	4.1519
			u_z	7.4620×10^{-11}	3.3650×10^{-12}	1.9973×10^{-13}	1.2351×10^{-14}	4.1869

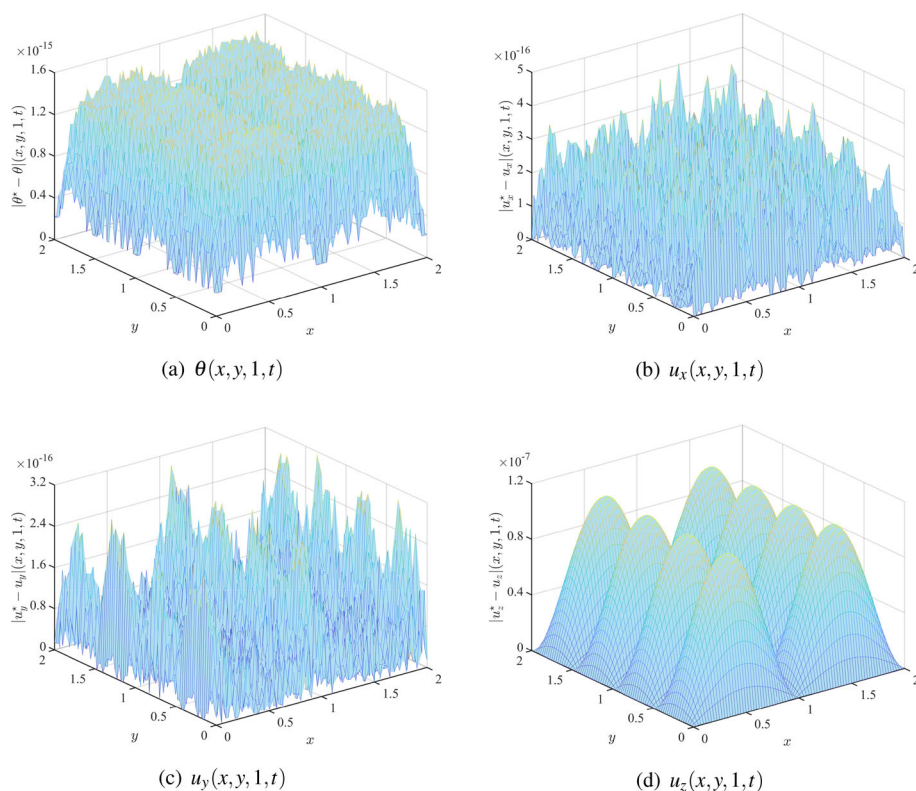


Fig. 6 The absolute errors between the analytical and numerical results of variables θ , u_x , u_y , and u_z ($\varepsilon = 0.2$, $t = 2.0$)

Maxwell iteration method. Furthermore, under the fifth-order approximation of the distribution function $\mathbf{f}$ in Eq. (23), we obtained the fourth-order MRT-LB model for the d -dimensional coupled Burgers' equations. It is worth mentioning that under the condition in Eq. (27), the gradient term $\nabla\theta$ in the d -dimensional diffusion equation can also be calculated locally by the non-equilibrium distribution function with a fourth-order accuracy. In addition, it should be noted that the developed fourth-order MRT-LB model for the coupled Burgers' equations is more general, and not limited to the spatial dimension. We also carried out some numerical experiments of the coupled Burgers' equations with homogeneous Dirichlet boundary conditions and periodic boundary conditions to test the developed MRT-LB model, and the numerical results show that the proposed MRT-LB model has a fourth-order convergence rate in space, which is also consistent with the theoretical analysis. Finally, we would also like to point out that in the framework of standard LB method, some high-order LB models can be developed for the linear diffusion equations and nonlinear coupled Burgers' equations, however, it is still difficult to generalize these high-order LB models to the problems with more than a quadratic non-linearity.

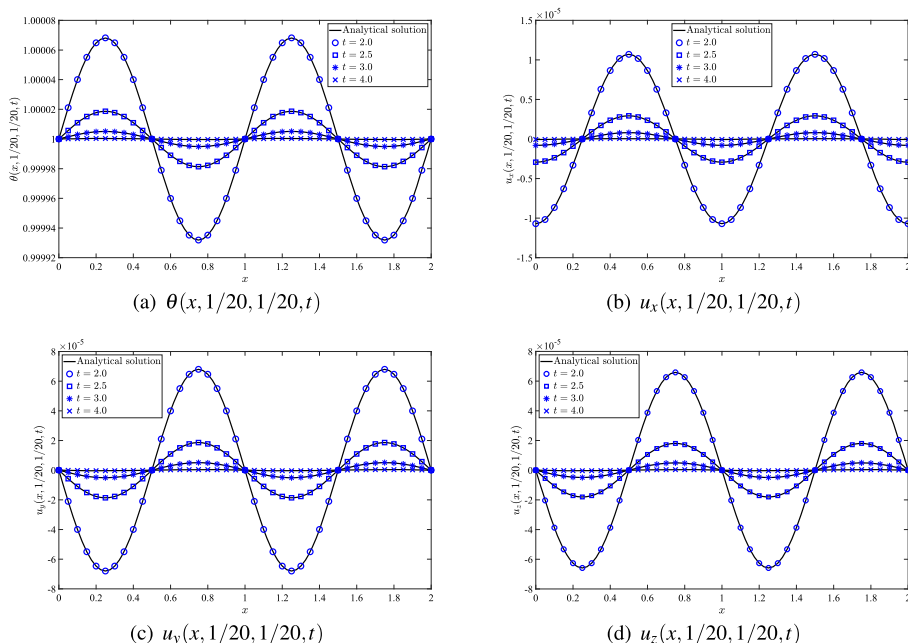


Fig. 7 Profiles of the variables θ , u_x , u_y , and u_z at different values of the time ($\varepsilon = 0.2$) (solid lines: analytical solutions, symbols: numerical results)

Appendix

A The expressions of $\mathbf{c}$, $\mathbf{p_M}(X_1, X_2, \dots, X_d)$, and $\mathbf{S}$ for the case $d > 3$

$$\mathbf{c} = \begin{pmatrix} Q_1, \underbrace{J_1^{d-1}, O_{\eta_1}}_{4(d-1)}, \\ Q_2, \underbrace{J_2, O_{4(d-2)}, J_1^{d-2}, O_{\eta_2}}_{4(d-1)}, \\ \vdots \\ Q_k, \underbrace{O_{4(k-2)}, J_2, O_{4(d-k)}, O_{4(k-3)}, J_2, O_{4(d-k)}, \dots, O_{4(k-l-1)}, J_2, O_{4(d-k)}, \dots, J_1^{d-k}, O_{\eta_k}}_{4(d-1)}, \\ \vdots \\ Q_d, \underbrace{O_{4(d-2)}, J_2, O_{4(d-3)}, J_2, \dots, O_{4(d-j-1)}, J_2, \dots, O_{4, J_2, J_2}}_{4(d-1)}, \underbrace{O_{4, J_2, J_2}}_{4 \times 2} \end{pmatrix}^c, \quad (30)$$

$$\mathbf{p_M}(X_1, X_2, \dots, X_d) = (1, X_1, X_2, \dots, X_d, X_1^2, X_2^2, \dots, X_d^2,$$

$$X_1 X_2, X_1 X_3, \dots, X_1 X_d, X_2 X_3, X_2 X_4, \dots, X_2 X_d, \dots, X_{d-1} X_d$$

$$X_1 X_2^2, X_1^2 X_2, \dots, X_1 X_d^2, X_1^2 X_d, X_2 X_3^2, X_2^2 X_3, \dots, X_2 X_d^2, X_2^2 X_d, \dots, X_{d-1} X_d^2, X_{d-1}^2 X_d,$$

$$X_1^2 X_2^2, X_1^2 X_3^2, \dots, X_1^2 X_d^2, X_2^2 X_3^2, X_2^2 X_4^2, \dots, X_2^2 X_d^2, \dots, X_{d-1}^2 X_d^2),$$

$$\mathbf{S} = \text{diag}(s_0, s_1 I_d, s_{21} I_d, s_{22} I_{d(d-1)/2}, I_{3d(d-1)/2}),$$

where $l = 1, 2, \dots, k-1$ ($k = 1, 2, \dots, d$), $j = 1, 2, \dots, d-1$ and^{3,4}

$$\begin{aligned} I_p &= (1, 1, \dots, 1) \in \mathbb{R}^{1 \times p}, \quad O_p = (0, 0, \dots, 0) \in \mathbb{R}^{1 \times p}, \\ Q_p &= (O_p, \underbrace{1}_{(p+1)_{th}}, O_{d-1}, \underbrace{-1}_{(d+p+1)_{th}}, O_{d-p}) \in \mathbb{R}^{1 \times (2d+1)}, \\ J_1 &= (1, -1, -1, 1), \quad J_2 = (1, 1, -1, -1), \\ J_1^p &= (\underbrace{J_1, J_1, \dots, J_1}_{4p}) \in \mathbb{R}^{1 \times (4p)}, \quad p = 1, 2, \dots, d-1, \\ \eta_i &= 2d(d-1) - 2i(2d-1-i). \end{aligned}$$

B The derivation of Eq. (29) from Eq. (28)

We first introduce the following two sets of polynomials,

$$\mathbf{r}_1(X_1, X_2, \dots, X_d) = (X_1, X_2, \dots, X_d), \quad (31a)$$

$$\mathbf{r}_2(X_1, X_2, \dots, X_d) = (X_1, X_2, \dots, X_1, X_d, X_2, X_3, \dots, X_2, X_d, \dots, X_{d-1}, X_d). \quad (31b)$$

With the aid of Eqs. (8) and (9), Eq. (28) can be simplified as

$$\begin{aligned} f_i(\mathbf{x}_w, t) - f_i(\mathbf{x}_w, t) &= -\Delta x (\Lambda_{ik}^{-1} - \Lambda_{ik}^{-1})(\mathbf{e}_k \cdot \nabla f_k^{eq}(\mathbf{x}_w, t)) \\ &= -\Delta x (\mathbf{M}_{ik}^{-1} - \mathbf{M}_{ik}^{-1})((\mathbf{S}^{-1}\mathbf{M})_{kl}\mathbf{e}_l \cdot \nabla f_l^{eq}(\mathbf{x}_w, t)). \end{aligned} \quad (32)$$

Based on the expressions of $\mathbf{S}$ and $\mathbf{p}_M$ in Eq. (30), we have

$$(\mathbf{S}^{-1}\mathbf{M})_{kl}(\mathbf{e}_l \cdot \nabla f_l^{eq}) = \left(0, \frac{c_s^2}{s_1 c} \mathbf{r}_1 \phi, O_{d(d+1)/2}, 4c^3(d-1)w_{2d+1}\mathbf{r}_2 \phi, O_{d(d-1)/2}\right)^T \quad (33)$$

where the element X_i of the polynomial set $\mathbf{r}_l$ ($l = 1, 2$) in Eq. (31) is taken as ∂_{x_i} , and the first- to fourth-order moments of the weight coefficients have been used, i.e.,

$$\begin{aligned} \sum_{i=1}^q w_i \mathbf{c}_{i\alpha} &= 0, \quad \sum_{i=1}^q w_i \mathbf{c}_{i\alpha} \mathbf{c}_{i\beta} = c_s^2 \delta_{\alpha\beta}, \quad \sum_{i=1}^q w_i \mathbf{c}_{i\alpha} \mathbf{c}_{i\beta} \mathbf{c}_{i\gamma} = 0, \\ \sum_{i=1}^q w_i \mathbf{c}_{i\alpha} \mathbf{c}_{i\beta} \mathbf{c}_{i\gamma} \mathbf{c}_{i\zeta} &= \begin{cases} c_s^2 c_s^2, & \alpha = \beta = \gamma = \zeta, \\ 4c^4(d-1)w_{2d+1}, & \alpha = \beta, \gamma = \zeta, \alpha \neq \gamma, \\ 0, & \text{others.} \end{cases} \end{aligned}$$

In the following, the derivation of Eq. (29) can be divided into two cases.

Case 1 The unknown distribution function f_i ($i = 2, 3, \dots, d+1$) at the boundary point $\mathbf{x}_w = \chi_{i-1}$ ($\chi = a_{i-1}$) which satisfies $\mathbf{x}_w - \mathbf{c}_i \in \Omega$ ($\hat{i} = d+i$) while $\mathbf{x}_w - \mathbf{c}_i \notin \Omega$.

³ The notation $\underbrace{J}_p$ represents the number of elements in the vector J being equal to p .

⁴ The notation $Q = (\dots, \underbrace{J}_{p_{th}}, \dots)$ represents J being the p_{th} element in the vector Q .

In this case, with the homogeneous boundary condition given by Eq. (5b), we have $\partial_{x_{i-1}}\phi(\mathbf{x}_w, t) = 0$, then Eq. (33) becomes

$$(\mathbf{S}^{-1}\mathbf{M})_{kl}(\mathbf{e}_l \cdot \nabla f_l^{eq}(\mathbf{x}_w, t)) = \left(0, \frac{c_s^2}{s_1 c} \mathbf{r}_1 \phi, O_{d(d+1)/2}, 4c^3(d-1)w_{2d+1}\mathbf{r}_2 \phi, O_{d(d-1)/2}\right)^T, \quad (34)$$

where the elements $X_{i-1} = 0$ and $X_j = \partial_{x_j}$ ($j = 1, 2, \dots, d, j \neq i-1$) of $\mathbf{r}_l$ ($l = 1, 2$) in Eq. (31). The term $(\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k}$ in Eq. (32) can be rewritten as

$$(\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k} = \left((O_{i-1}, 1, O_{d-1}, -1, 0, \dots, 0)\mathbf{M}^{-1}\right)_k,$$

and from the expression of the matrix $\mathbf{M}^{-1}$ presented in Ref. [17], the above equation becomes

$$\begin{aligned} (\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k} &= (O_{i-1}, 1, O_{d-1}, -1, 0, \dots, 0)\mathbf{M}^{-1} \\ &= \frac{1}{c^3}(0, c^2\mathbf{r}_1, O_{d(d+1)/2}, -\mathbf{r}_2, O_{d(d-1)/2}), \end{aligned} \quad (35)$$

where the elements $X_{i-1} = 1$ and $X_j = 0$ ($j = 1, 2, \dots, d, j \neq i-1$) of $\mathbf{r}_l$ ($l = 1, 2$) in Eq. (31). Thus, according to Eqs. (34) and (35), we have

$$(\mathbf{M}_{ik}^{-1} - \mathbf{M}_{\hat{i}k}^{-1})(\mathbf{S}^{-1}\mathbf{M})_{kl}\mathbf{e}_l \cdot \nabla f_l^{eq}(\mathbf{x}_w, t) = 0,$$

and further obtain Eq. (29) for the case $i = 2, 3, \dots, d+1$. We would like to point out that the case of the unknown distribution function $f_i(\chi_{\hat{i}-1}, t)$ with $\chi = b_{\hat{i}-1}$ ($i = d+2, \dots, 2d+1$) can be analyzed similarly, and we do not present the details here.

Case 2 The unknown distribution function f_i ($i = 2d+2, 2d+3, \dots, q$) at the boundary $\partial\Omega$.

Here we take $i = 2d+2$ as an example, which corresponds x_1 and x_2 coordinate directions. From Eq. (5b), we have

$$\begin{cases} \partial_{x_1}\phi(\chi_1, t) = 0 \text{ with } \chi = a_1 \text{ and } \hat{i} = i+1, \\ \partial_{x_2}\phi(\chi_2, t) = 0 \text{ with } \chi = a_2 \text{ and } \hat{i} = i+3, \end{cases}$$

then Eq. (33) becomes

$$\begin{aligned} (\mathbf{S}^{-1}\mathbf{M})_{kl}(\mathbf{e}_l \cdot \nabla f_l^{eq}(\chi_1, t)) &= \left(0, \frac{c_s^2}{s_1 c} \mathbf{r}_1(0, \partial_{x_2}, \dots, \partial_{x_d})\phi, O_{d(d+1)/2}, 4c^3(d-1)w_{2d+1}\mathbf{r}_2 \right. \\ &\quad \left. (0, \partial_{x_2}, \dots, \partial_{x_d})\phi, O_{d(d-1)/2}\right)^T, \\ (\mathbf{S}^{-1}\mathbf{M})_{kl}(\mathbf{e}_l \cdot \nabla f_l^{eq}(\chi_2, t)) &= \left(0, \frac{c_s^2}{s_1 c} \mathbf{r}_1(\partial_{x_1}, 0, \partial_{x_3}, \dots, \partial_{x_d})\phi, O_{d(d+1)/2}4c^3(d-1)w_{2d+1}\mathbf{r}_2 \right. \\ &\quad \left. (\partial_{x_1}, 0, \partial_{x_3}, \dots, \partial_{x_d})\phi, O_{d(d-1)/2}\right)^T. \end{aligned}$$

For the term $(\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k}$ in Eq. (32), similar to the Case 1, we can obtain

$$\begin{aligned} \hat{i} = i+1 : (\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k} &= \left((O_{1+2d}, 1, -1, 0, \dots, 0)\mathbf{M}^{-1}\right)_k \\ &= \left(\frac{1}{2c^3}(O_{1+2d}, c, O_{d^2-d-2}/2, 1, 0, \dots, 0)\right)_k, \\ \hat{i} = i+3 : (\mathbf{M}^{-1})_{ik} - (\mathbf{M}^{-1})_{\hat{i}k} &= \left((O_{1+2d}, 1, -1, 0, \dots, 0)\mathbf{M}^{-1}\right)_k \end{aligned}$$

$$= \left(\frac{1}{2c^3} (O_{1+2d}, c, O_{(d^2-d)/2}, 1, 0, \dots, 0) \right)_k.$$

And it is easy to show that

$$\begin{aligned} (\mathbf{M}_{ik}^{-1} - \mathbf{M}_{ik}^{-1})((\mathbf{S}^{-1}\mathbf{M})_{kl}\mathbf{e}_l \cdot \nabla f_l^{eq}(\chi_1, t)) &= 0, \\ (\mathbf{M}_{ik}^{-1} - \mathbf{M}_{ik}^{-1})((\mathbf{S}^{-1}\mathbf{M})_{kl}\mathbf{e}_l \cdot \nabla f_l^{eq}(\chi_2, t)) &= 0. \end{aligned}$$

Actually, the analysis on the other cases with $i = 2d + 3, 2d + 4, \dots$, q is similar, and the details are not shown here. Based on the results of Case 1 and Case 2, we can obtain Eq. (29).

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Declarations

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